

Quantum Computing Research Portfolio

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github.com/asmeeta-quantum/ETH_Qiskit_Workspace

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Abstract

This portfolio synthesizes my independent research in quantum information theory, focusing on how realistic noise processes influence computational power, verifiability, and classical simulability boundaries. My work combines analytical modeling of open quantum systems with numerical simulations using Qiskit and QuTiP. Background at ISRO provides practical insight into physical noise sources in cryogenic and microwave systems relevant for quantum devices.

Research Goal: Develop rigorous frameworks for verification complexity, noise-adapted quantum advantage thresholds, and resource-theoretic descriptions of noisy quantum computation.

Research Alignment with ETH Zurich

My research aligns with the Quantum Information Group at ETH Zurich, particularly the work of:

- **Prof. Renato Renner:** Quantum information theory, smooth entropies, verification complexity
- **Prof. Juan Carrasquilla:** Quantum machine learning, quantum many-body physics
- **Prof. David Steurer:** Complexity theory, hardness of approximation
- **Prof. Matthias Troyer:** Quantum advantage, computational limits of near-term devices

Key Research Contributions

Verification Complexity Under Noise

Objective: Understand how T_1/T_2 relaxation and dephasing affect verification sample complexity.

Methods:

- Analytical scaling bounds for verification under noise channels
- Quantum channel divergences and entropy measures
- Identification of noise thresholds for classical simulability

Key Result: Derived analytical bounds identifying regimes where quantum advantage becomes classically simulable.

Error Mitigation Theory and Simulation

Objective: Develop and validate error mitigation frameworks for noisy quantum circuits.

Methods:

- Zero-Noise Extrapolation (ZNE) implementation
- Systematic noise amplification analysis
- Benchmarking on simulated noisy circuits

Key Result: Demonstrated $\approx 20\%$ fidelity improvement on noisy quantum circuits.

Quantum Control Optimization

Objective: Suppress non-computational excitations in weakly anharmonic transmon qubits.

Methods:

- DRAG pulse optimization framework

- Lindblad master equation simulations
- Leakage-aware control theory

Key Result: Achieved > 90% suppression of $|1\rangle \rightarrow |2\rangle$ leakage.

Hardware-Informed Noise Modeling

Objective: Translate microwave component stability data into quantum noise parameters.

Methods:

- Phase noise characterization from VNA measurements
- Cryogenic system stability analysis (4K)
- Composite noise channel modeling

Key Insight: Hardware characterization data provides practical constraints for quantum noise models, bridging theory and experiment.

Research Portfolio Structure

```
ETH_Qiskit_Workspace/  
01_bell_states_entanglement.ipynb  
02_statevector_simulation.ipynb  
03_depolarizing_noise_analysis.ipynb  
04_superconducting_noise_channels.ipynb (T1/T2 modeling for transmons)  
05_composite_noise_simulation.ipynb  
06_grover_verification.ipynb  
07_deutsch_jozsa_advantage.ipynb  
08_verification_sample_complexity.ipynb (Key result: analytical bounds)  
09_lindblad_qubit_dynamics.ipynb  
10_drag_pulse_optimization.ipynb (Key result: > 90% suppression)  
11_noise_bench_qc/ # Benchmarking suite  
12_lindblad_sim/ # Simulation toolkit
```

Technical Implementation

Software Tools Developed

Tool	Description	Key Result
Noise-Bench QC	Benchmarking suite for verification under realistic noise	Systematic noise characterization framework
Lindblad-Sim	Non-Markovian and leakage-aware simulation toolkit	Open system dynamics simulation
ZNE Toolkit	Zero-Noise Extrapolation framework	≈ 20% fidelity improvement
DRAG Pulse Optimizer	Adaptive pulse design for leakage suppression	> 90% leakage suppression

Mathematical Methods

- **Smooth Entropies:** Quantifying information leakage and verification complexity
- **Quantum Channel Divergences:** Characterizing recoverability and noise thresholds
- **Lindblad Dynamics:** Modeling composite noise evolution
- **Resource Theories:** Formalizing noisy quantum advantage

Technical Skills

Quantum Theory	Open quantum systems, Lindblad dynamics, quantum channels, verification protocols, entropy measures, resource theories
Quantum Software	Qiskit (Terra, Aer, Pulse), QuTiP, OpenQASM
Programming	Python (NumPy, SciPy, Matplotlib), Git, LaTeX
Hardware	Cryogenic systems (4K), microwave instrumentation, VNA measurements, phase noise characterization

Professional Experience Integration

Space Applications Centre (ISRO) — Senior Assistant

2014–Present

- Evaluate cryogenic and microwave components for space payloads operating near 4K
- Analyze phase noise, amplifier noise figures, and S-parameter measurements from VNA characterization reports
- Interpret hardware noise characteristics and connect experimental parameters to abstract quantum noise channel models
- Bridge hardware characterization with quantum noise modeling

Research Contributions Summary

1. **Analytical bounds** linking physical noise (T_1/T_2) to verification scalability
2. **Open-source tools** for realistic quantum noise simulation
3. **Hardware-to-quantum modeling pipeline** from ISRO experience
4. **Identification of noise-triggered simulability transitions**
5. **> 90% leakage suppression** via DRAG pulse optimization
6. **$\approx 20\%$ fidelity improvement** via zero-noise extrapolation

Proposed PhD Research Agenda

Planned research directions under ETH Zurich supervision:

- **Noise-adapted verification protocols:** Extend analytical bounds to broader circuit classes
- **Resource thresholds for genuine quantum advantage:** Quantify conditions for practical advantage
- **Calibration-efficient experimental certification:** Connect theory to NISQ device constraints
- **Classical simulability boundaries:** Map noise thresholds for quantum-to-classical transitions

Why ETH Zurich

ETH Zurich uniquely enables research at the intersection of:

- **Theoretical foundations** (Renner, Hangleiter)
- **Quantum machine learning** (Carrasquilla)
- **Complexity theory** (Steurer)
- **Computational quantum advantage** (Troyer)

This combination provides the ideal environment for developing rigorous frameworks for noise-adapted quantum computation.

Repository and Code Availability

Full Repository: https://github.com/asmeeta-quantum/ETH_Qiskit_Workspace

All notebooks, simulation toolkits, and benchmarking suites are available for review.

Contact: asmitasayaji21@gmail.com

GitHub: github.com/asmeeta-quantum/ETH_Qiskit_Workspace